

Clinical Perspective

Nutrition interventions for adults with dyslipidemia: A Clinical Perspective from the National Lipid Association[☆]



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KEY WORDS

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Abstract: Lifestyle habits can have a profound impact on atherosclerotic cardiovascular disease (ASCVD) risk. The National Lipid Association previously published recommendations for lifestyle therapies to manage dyslipidemia. This Clinical Perspective provides an update with a focus on nutrition interventions for the three most common dyslipidemias in adults: 1) low-density lipoprotein cholesterol (LDL-C) elevation; 2) triglyceride (TG) elevation, including severe hypertriglyceridemia with chylomicronemia; and 3) combined dyslipidemia, with elevations in both LDL-C and TG levels. Lowering LDL-C and non-high-density lipoprotein cholesterol are the primary objectives for reducing ASCVD risk. With severe TG elevation (≥ 500 mg/dL), the primary objective is to prevent pancreatitis and ASCVD risk reduction is secondary. Nutrition interventions that lower LDL-C levels include reducing cholesterol-raising fatty acids and dietary cholesterol, as well as increasing intakes of unsaturated fatty acids, plant proteins, vis-

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cous fibers, and reducing adiposity for patients with overweight or obesity. Selected dietary supplements may be employed as dietary adjuncts. Nutrition interventions for all patients with elevated TG levels include restricting intakes of alcohol, added sugars, and refined starches. Additional lifestyle factors that reduce TG levels are participating in daily physical activity and reducing adiposity in patients with overweight or obesity. For patients with severe hypertriglyceridemia, an individualized approach is essential. Nutrition interventions for addressing concurrent elevations in LDL-C and TG include a combination of the strategies described for lowering LDL-C and TG. A multidisciplinary approach is recommended to facilitate success in making and sustaining dietary changes and the assistance of a registered dietitian nutritionist is highly recommended.

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Introduction

In 2014-2015, the National Lipid Association (NLA) published Recommendations for Patient-Centered Management of Dyslipidemia Parts 1 and 2 to provide in-depth clinical guidance for the assessment and treatment of dyslipidemias in adults.^{1,2} Since then, other scientific organizations have published guidelines and recommendations for the prevention and management of atherosclerotic cardiovascular disease (ASCVD).³⁻⁸ However, nutrition recommendations in those documents typically address ASCVD prevention, in general, rather than nutrition interventions for the management of specific dyslipidemias.

The NLA Recommendations-Part 2 included both comprehensive dietary and other lifestyle recommendations. In 2022, the NLA published a Clinical Perspective for health-care professionals that described nutrition interventions for youth (<18 years) with disorders of lipid and lipoprotein metabolism.⁹ In this 2023 NLA Clinical Perspective, detailed guidance is provided on nutrition interventions for dyslipidemia management in adults (≥18 years). The focus of this Clinical Perspective is on mitigating ASCVD risk and other adverse clinical outcomes (e.g., pancreatitis, eruptive and tendon xanthomas) through the dietary management of the three most common dyslipidemias: 1) isolated low-density lipoprotein cholesterol (LDL-C) elevation; 2) isolated triglyceride (TG) elevation, including chylomicronemia; and 3) combined dyslipidemia, including elevations in both LDL-C and TG levels. In a majority of patients, the key factor driving ASCVD risk that is modifiable through dietary intervention is the number of circulating atherogenic lipoprotein particles (i.e., apolipoprotein B [Apo B]-containing particles), for which LDL-C and non-HDL-C are clinically measured surrogates.^{10,11} Although other dyslipidemias, such as low levels of high-density lipoprotein cholesterol (HDL-C) and elevated lipoprotein(a) [Lp(a)], are commonly encountered in clinical practice, evidence is lacking to demonstrate clinical value for dietary interventions for these dyslipidemias.

Dietary habits, lifestyle, other determinants, and ASCVD risk

A healthy lifestyle is the foundation of ASCVD prevention,³⁻⁸ and good dietary habits are fundamental to a healthy lifestyle. Evidence from the Global Burden of Disease Study 2019 suggested that, in the United States (U.S.), adverse dietary habits are among the leading contributing factors for loss of disability free life years.^{12,13} Additional modifiable risk factors for ASCVD risk include physical inactivity, inadequate sleep quantity and quality, smoking/tobacco use, elevated blood pressure, hyperglycemia (prediabetes or diabetes), chronic inflammation, and excess adiposity.^{4,5,7,8,14,15} Notably, dietary behaviors are intricately connected with other lifestyle-related ASCVD risk factors, all of which may be impacted by social determinants of health, including economic stability, education quality, as well as access to nutritious foods and safe environments for exercise. Importantly, improving poor dietary habits, which are disproportionately present in economically disadvantaged and racial/ethnic minority groups, is aligned with commitments by the NLA, the American Heart Association (AHA), and other scientific organizations, to improve cardiovascular (CV) health equity.^{16,17} Thus, although this Clinical Perspective focuses on the influences of nutrition interventions on lipoprotein lipid levels, diet-induced improvements in other risk factors (e.g., hypertension, hyperglycemia, and chronic inflammation) will also contribute to ASCVD risk reduction.^{1,2,4-8}

The AHA has summarized key metrics for CV health in a Presidential Advisory titled *Life's Essential Eight*.¹⁸ This document outlines the health behaviors and metrics associated with optimal CV health (Table 1). Numerous studies have demonstrated that achieving and maintaining these health behaviors and favorable levels of these key metrics is associated with substantially lower incidence of ASCVD, as well as lower risks for several types of cancer, dementia, and other chronic diseases of aging, and also with improved quality of life and greater healthy longevity.¹⁸

Table 1 Life's Essential Eight¹⁸

Behaviors	Metrics
Healthy eating pattern – optimal score achieved with high adherence to Mediterranean- or DASH-style eating pattern	BMI – optimal score achieved with BMI <25 kg/m ²
Physical activity – optimal score achieved with ≥150 minutes/week of moderate or 75 minutes/week of vigorous physical activity	Blood lipids – optimal score achieved with non-HDL-C <130 mg/dL
Nicotine exposure – optimal score achieved with self-reported “never smoker”; encourage patients to seek assistance for cessation, as needed	Fasting blood glucose and HbA1c – optimal score achieved with no history of diabetes and FBG <100 mg/dL or HbA1c <5.7%
Sleep health – optimal score achieved with self-reported 7 to <9 hours of sleep each night	Blood pressure – optimal score achieved with systolic and diastolic blood pressure <120/<80 mm Hg, respectively
Abbreviations: ASCVD=atherosclerotic cardiovascular disease; BMI=body mass index; DASH=Dietary Approaches to Stop Hypertension; FBG=fasting blood glucose; HbA1c=hemoglobin A1c; non-HDL-C=non-high-density lipoprotein cholesterol	

Dietary patterns associated with reduced ASCVD risk

A healthy dietary pattern beneficially impacts atherogenic lipoprotein levels, as well as other ASCVD risk factors.^{4-8,19} The foods and components of a healthy dietary pattern can mitigate ASCVD risk by favorably affecting numerous metabolic pathways and cellular processes, including those that affect the atherothrombotic process, such as atherosclerotic plaque initiation and progression, plaque instability and rupture, and thrombosis,²⁰ although the mechanistic pathways are only partially understood.²¹ Emerging research on the influences of dietary patterns and food components on metabolomics, lipidomics, and the microbiome may increase the understanding of the complex interaction of diet and ASCVD risk.^{21,22} Thus, nutrition interventions for specific dyslipidemias should be incorporated into an overall healthy dietary pattern.

Healthy dietary patterns provide cumulative exposures to combinations of foods, nutrients, and bioactive compounds associated with cardiometabolic health benefits when consumed at appropriate levels of energy intake,²³⁻²⁵ as well as minimize exposure to harmful dietary components. Consequently, they provide greater CV and other health benefits than typically achieved with single nutrient substitutions, which is why healthy eating patterns are emphasized in contemporary dietary recommendations.^{4-8,19} Several healthy dietary patterns have been described in recent Dietary Guidelines for Americans¹⁹ and are included in nutrition recommendations from scientific organizations, such as the AHA,⁶ European Society of Cardiology,^{5,8} and European Atherosclerosis Society.⁵ The recommended dietary patterns include the Mediterranean, Dietary Approaches to Stop Hypertension (DASH), healthy vegetarian (including vegan), and the Healthy U.S.-Style dietary patterns.

Each dietary pattern has distinct characteristics but all share commonalities (Fig. 1), including an emphasis on whole, unprocessed, or minimally processed, foods with high intakes of fiber-rich components, such as fruits and vegetables, whole grains, nuts, and seeds. These dietary patterns

also emphasize selection of healthy protein sources, such as fish and seafood, plant-based proteins (e.g., nuts, seeds, legumes/pulses), and lower fat animal proteins (meats, poultry, dairy, eggs), as well as non-tropical, liquid plant oils in place of solid (more saturated) fats. Healthy dietary patterns are low in highly processed foods, including processed meats, refined grains, and foods/beverages with added sugars and salt, and are low in foods with high contents of saturated fatty acids (SFAs) and *trans* fatty acids (TFAs).^{4-8,19,26} High diet quality, which is associated with reduced ASCVD risk, is emphasized in each of the recommended dietary patterns.²⁷ All individuals should be informed about how to implement a healthy eating pattern that aligns with their personal preferences, budget constraints, and cultural traditions. For individuals who have dyslipidemia, personalized counseling is important and recommended.^{4-6,8,28} Personalized medical nutrition therapy for dyslipidemia management provided by registered dietitian nutritionists (RDNs) is clinically beneficial and cost effective.²⁹⁻³¹ Patient education handouts for the recommended healthy dietary patterns and specific nutrition interventions are available on the NLA Clinician's Lifestyle Modification Toolbox webpage (www.lipid.org/clmt).

Lipoprotein lipid targets of nutrition interventions and other lifestyle therapies

The targets of nutrition and other lifestyle interventions for ASCVD risk reduction are mainly circulating levels of atherogenic particles, i.e., Apo B-containing lipoproteins. Since each potentially atherogenic lipoprotein particle [LDL, which includes intermediate density lipoprotein (IDL), very-low-density lipoprotein (VLDL), chylomicron remnants, and Lp(a)] contains a single molecule of Apo B, the concentration of Apo B is a direct indication of the circulating burden of atherogenic particles.¹¹ Because Apo B is not routinely measured clinically in the U.S., both LDL-C and non-HDL-C are measured surrogates for Apo B-containing lipoproteins. The non-HDL-C concentration is comprised of cholesterol carried by all Apo B-containing lipoprotein particles and correlates more strongly than LDL-C with the Apo

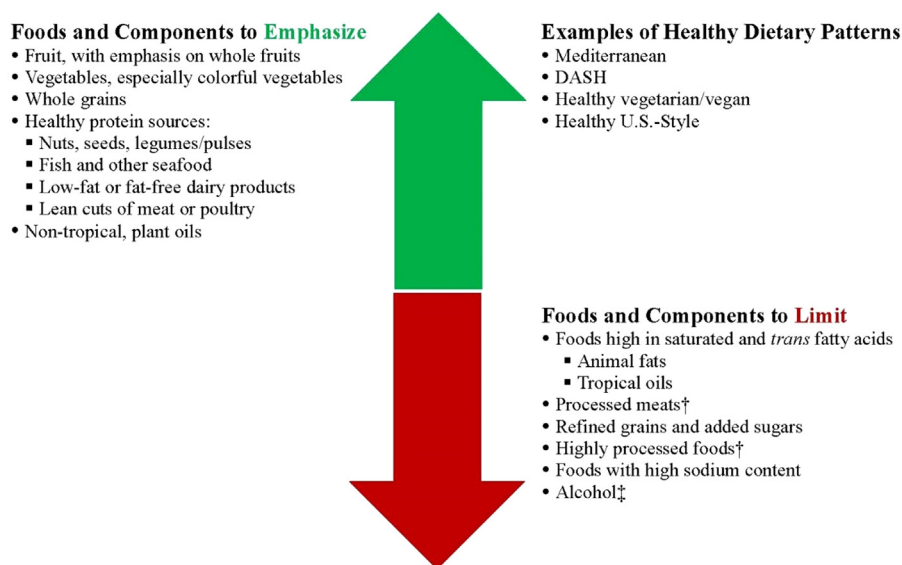


Fig. 1 Commonalities of dietary patterns associated with reduced ASCVD risk*4-8,19,26

*Although healthy dietary patterns share commonalities, they emphasize different components, e.g., DASH has a greater emphasis on fruits, vegetables, and low-fat dairy and the Mediterranean dietary pattern has a greater emphasis on a higher intake of foods rich in MUFAs, such as nuts and extra-virgin olive and canola oils. Foods and dietary components to emphasize or limit is recommended regardless of where food is prepared or consumed.

†Examples of processed meats to limit include bacon, sausage, salami, ham, pepperoni, and deli meats. Examples of highly processed foods to limit include ready-to-eat and quick service meals, refined breads and baked goods, fruit juices, commercially prepared foods and beverages with added sugar or salt, sauces, and condiments.

‡If individuals choose to drink alcohol, intake should be moderate: ≤ 1 drink/day for women; ≤ 2 drinks/day for men. 1 drink = 12 fl. oz regular beer, 8-10 fl. oz malt beverage, 5 fl. oz table wine, 3-4 fl. oz fortified wine, 2-3 fl. oz liqueur, 1.5 fl. oz brandy/cognac, or 1.5 fl. oz distilled spirit.

Abbreviations: ASCVD=atherosclerotic cardiovascular disease; DASH=Dietary Approaches to Stop Hypertension; MUFAs=monounsaturated fatty acids

B concentration, particularly in patients with elevated TG and/or low HDL-C, thus is an important target for intervention.^{1,2,5,7,8,32} However, since LDL-C is the cholesterol fraction carried by the most abundant atherogenic particle in most individuals and has been the stated target of therapy for many intervention studies, it is generally considered the primary target of interventions aimed at reducing ASCVD risk.¹⁰ An exception is dysbetalipoproteinemia (type III dyslipidemia), which is associated with increased ASCVD risk even when the true LDL-C and Apo B levels are not elevated. Therefore, non-HDL-C is the preferred target of therapy in dysbetalipoproteinemia because it incorporates the cholesterol carried by remnant (i.e., chylomicron, VLDL, and IDL) particles. Dysbetalipoproteinemia is discussed in more detail in the “Elevation in LDL-C and TG levels (mixed dyslipidemia)” section under “Categories of Dyslipidemias.”

The primary objective for lowering atherogenic lipoprotein lipid (LDL-C and non-HDL-C) levels is to reduce the risk of a first CV event in primary prevention or future CV events in secondary prevention. Although lifestyle interventions are the foundation for all individuals with elevated LDL-C and non-HDL-C levels, many patients also require pharmacotherapy.^{3,5,7,8} Table 2 summarizes therapeutic objectives for patients at different levels of ASCVD risk as defined by the 2018 AHA/American College of Cardiology (ACC)/Multisociety Guideline on the Management of Blood

Cholesterol³ and the 2022 ACC Expert Consensus Decision Pathway on the Role of Nonstatin Therapies for LDL-C Lowering in the Management of ASCVD Risk,³³ which are generally aligned with those in European⁸ and Canadian guidelines⁷.

Limited evidence is available from randomized, controlled trials (RCTs) from which to directly assess the impacts of nutrition interventions to lower LDL-C and non-HDL-C on ASCVD event risk. However, results from RCTs, epidemiologic, and genetic (Mendelian randomization) studies consistently support a strong positive relationship between circulating levels of LDL-C (as well as non-HDL-C) and ASCVD event risk.^{1,2,34} Results from the Cholesterol Treatment Trialists’ collaboration analysis demonstrated that each 1 mmol/L (38.7 mg/dL) reduction in LDL-C with statin therapy is associated with an average 22% reduction in major vascular event risk over a period of ~ 5 years, with similar reductions in risks for subcategories of events (e.g., myocardial infarction, ischemic stroke, and revascularization).³⁵ Evidence from observational studies with an average follow-up of ~ 12 years demonstrated that a 1 mmol/L lower concentration of LDL-C is associated with $\sim 32\%$ lower coronary heart disease event risk, whereas genetic variants that reduce LDL-C levels over multiple decades are associated with a 50-55% lower risk per mmol/L reduction.³⁴ Thus, relatively small reductions in LDL-C have the potential to sub-

Table 2 Therapeutic objectives for LDL-C and non-HDL-C by ASCVD risk category^{3,33}

ASCVD Risk Category	LDL-C (mg/dL)	Non-HDL-C (mg/dL)
Low risk ^a	<100	<130
Borderline to intermediate risk ^b	<100	<130
High risk ^c	<70	<100
Very high risk ^{d,e}	<55	<85

^aLow risk refers to an estimated 10-year risk for ASCVD of <5%.

^bBorderline risk refers to an estimated 10-year risk for ASCVD of 5% to <7.5%. Intermediate risk refers to the presence of severe primary hypercholesterolemia (LDL-C $\geq$ 190 mg/dL) or an estimated 10-year risk for ASCVD of 7.5% to <20%.

^cHigh risk refers to the presence of clinical ASCVD with or without severe hypercholesterolemia (LDL-C $\geq$ 190 mg/dL), diabetes mellitus, or an estimated 10-year risk for ASCVD of $\geq$ 20%.

^dVery high risk includes a history of multiple major ASCVD events (e.g., recent ACS, MI, ischemic stroke, symptomatic PAD) or 1 major ASCVD event and multiple high-risk conditions (e.g., $\geq$ 65 years of age, heterozygous FH, history of CABG or PCI, diabetes mellitus, HTN, CKD, current smoking, persistently elevated LDL-C $\geq$ 100 mg/dL despite statin + ezetimibe, history of HF).

^eAn optional therapeutic objective of LDL-C <40 mg/dL may be considered in patients with ASCVD who experience a second vascular event within 2 years while taking maximally tolerated statin therapy.⁵

Abbreviations: ACS=acute coronary syndrome; ASCVD=atherosclerotic cardiovascular disease; CABG=coronary artery bypass graft; CKD=chronic kidney disease; FH=familial hypercholesterolemia; HF=heart failure; HTN=hypertension; LDL-C=low-density lipoprotein cholesterol; MI=myocardial infarction; non-HDL-C=non-high-density lipoprotein cholesterol; PAD=peripheral artery disease; PCI=percutaneous coronary intervention

stantially reduce ASCVD risk if maintained over an extended period.

The 2018 AHA/ACC/Multisociety Guideline on the Management of Blood Cholesterol defined elevated TG levels (persistently $\geq$ 175mg/dL) as a risk-enhancing factor for primary ASCVD prevention.³ An elevated TG concentration is one of the components that defines the metabolic syndrome and is a strong indication for lifestyle interventions. The TG level *per se* is not a target of therapy, except when levels are very high ($\geq$ 500 mg/dL). The primary objective for individuals with TG levels $\geq$ 500 mg/dL is to achieve and maintain a TG concentration <500 mg/dL to reduce the risk for acute pancreatitis. This may be achieved through nutrition and other lifestyle interventions, as well as pharmacotherapy when appropriate. In such patients, a secondary objective is to manage levels of LDL-C and non-HDL-C, as well as other modifiable risk factors, to lower ASCVD risk. For those with mild-to-moderate hypertriglyceridemia, reducing ASCVD risk by attaining acceptable levels of LDL-C and non-HDL-C is the primary objective of intervention.^{1,3,5,7,8,32}

HDL-C is not a target of therapy, although low HDL-C is a strong predictor of ASCVD event risk and a factor in the Pooled Cohort Equation calculations used to estimate ASCVD risk.³ Nutrition interventions that lower levels of LDL-C and non-HDL-C often alter HDL-C, but the clinical relevance of these changes is uncertain at present.^{1,2,5,8}

Lp(a) is also not a target of nutrition interventions, although it is considered an independent and causal risk factor for ASCVD.^{9,36-39} An elevated Lp(a) level is a risk-enhancing factor, and an elevated level ($\geq$ 50 mg/dL or 125 nmol/L) often warrants intensifying the management of other ASCVD risk factors.^{1,3,5,7,36-39} At the time of the writing of this Clinical Perspective, trials were underway to assess the effects of pharmacologic interventions for Lp(a) reduction on ASCVD risk. There is limited and inconclusive evidence that dietary interventions alter Lp(a) levels.^{9,36-38} The few RCTs that have examined the effects of nutrition interventions on Lp(a) levels have found inconsistent results that are of uncertain clinical relevance.⁴⁰ Therefore, Lp(a) reduction will not be considered further in this Clinical Perspective.

Categories of dyslipidemias for targeted nutrition interventions

As discussed previously, the focus of this NLA Clinical Perspective is on three specific dyslipidemias: 1) isolated elevated LDL-C levels; 2) isolated elevated TG levels, including chylomicronemia; and 3) combined dyslipidemia, defined as elevation of both LDL-C and TG levels. Table 3 illustrates the characteristics of these dyslipidemias, including the lipoproteins and lipids affected, clinical findings, and, for historical reference, the Fredrickson-Levy-Lees phenotype(s) for each lipid disorder.

Isolated elevated LDL-C levels

An elevated LDL-C level has been the therapeutic target in blood cholesterol guidelines since the National Cholesterol Education Program: Adult Treatment Panel I was published in 1988 after evidence from early RCTs showed that reducing LDL-C was associated with lower risk for ASCVD events.⁴⁴

Pathophysiology of LDL-C elevation

Genetics play a key role in driving LDL-C elevation, although nutrition and other lifestyle factors have a substantial impact on the circulating LDL-C concentration. With polygenic hypercholesterolemia, there are multiple gene polymorphisms that mainly impair LDL clearance pathways.⁴⁵ In contrast, familial hypercholesterolemia (FH) is an autosomal dominant, monogenic disorder, which can result in severely elevated LDL-C levels.⁴⁵⁻⁴⁸ In FH, pathogenic genetic variants can cause LDL receptor deficiency or dysfunction, defective Apo B, or proprotein convertase subtilisin/kexin type 9 (PCSK9) gain-of-function.⁴⁵⁻⁴⁸ Heterozygous FH (HeFH) is typically caused by a monoallelic pathogenic variant (i.e., inherited from one parent) in one of the primary genes associated with FH (LDL receptor, including *LDLR*, *APOB*, *PCSK9*) and is among the most common inherited disorders of lipoprotein metabolism ($\sim$ 1 in 250 individuals).^{48,49} Patients with HeFH commonly present with LDL-C levels $\geq$ 190 mg/dL, although many will have lower levels, especially in young adulthood.^{3,8,48,50} Homozygous FH (HoFH) is rare ($\sim$ 1 in 160,000-300,000 individuals) and results from

Table 3 Categories of dyslipidemia⁴¹⁻⁴³

Dyslipidemia*	Lipoprotein(s) Affected	Lipid(s) Affected	Clinical Findings	Fredrickson- Levy-Lees Phenotype(s)
Isolated LDL-C elevation				
Familial hypercholesterolemia	↑ LDL	↑↑↑ LDL-C	Tendon xanthomas, xanthelasma, arcus cornealis; increased ASCVD risk	IIa, IIb
Polygenic hypercholesterolemia	↑ LDL	↑ to ↑↑↑ LDL-C	ASCVD risk	IIa
Isolated TG elevation				
Familial chylomicronemia syndrome	↑ Chylomicrons ↑ VLDL	↑↑↑ TG	Risk for pancreatitis, eruptive xanthomas (TG ≥2000 mg/dL), lipemia retinalis (TG ≥4000 mg/dL), hepatosplenomegaly, ASCVD risk not increased if Apo B is not elevated	I
Multifactorial hypertriglyceridemia	↑ Chylomicrons ↑ VLDL	↑ to ↑↑↑ TG	Risk for pancreatitis ASCVD risk if Apo B is elevated	IV, V
Mixed dyslipidemia (elevation in LDL-C and TG)				
Dysbetalipoproteinemia ("true" LDL-C is not typically elevated but IDL-C is generally elevated [†])	↑ Chylomicron remnants ↑ VLDL ↑ IDL	↑ to ↑↑↑ TG LDL-C may be normal to ↑↑ (esp. IDL-C)	Palmar xanthomas, ASCVD risk	III
Mixed dyslipidemia	↑ VLDL, ↑ IDL, ↑ LDL	↑ to ↑↑↑ LDL-C ↑↑ TG	ASCVD risk	IIb

*Dyslipidemias related to HDL-C and Lp(a) levels are not a focus of this Clinical Perspective and, therefore, not included in this table. HDL-C disturbances can occur, which result in decreased HDL and HDL-C concentrations. Low HDL-C is a strong predictor of ASCVD but not a target of therapies. An elevated Lp(a) level has mainly genetic determinants and is a causal risk-enhancing factor for ASCVD.

[†]When LDL-C is calculated, IDL-C is included in the estimate of LDL-C. Therefore, in patients with dysbetalipoproteinemia, "true" LDL-C levels are generally not elevated but may appear so on laboratory reports because of IDL-C elevation.

Abbreviations: Apo B=apolipoprotein B; ASCVD=atherosclerotic cardiovascular disease; HDL=high-density lipoprotein; HDL-C=high-density lipoprotein cholesterol; IDL=intermediate-density lipoprotein; IDL-C=intermediate-density lipoprotein cholesterol; LDL=low-density lipoprotein; LDL-C=low-density lipoprotein cholesterol; Lp(a)=lipoprotein(a); TG=triglyceride; VLDL=very-low-density lipoprotein.

inheriting biallelic pathogenic variants (i.e., one from each parent) in *LDLR*, *APOB*, *PCSK9*, and/or *LDLRAP*. Such individuals often have LDL-C levels ≥400 mg/dL.⁴⁵⁻⁴⁷ Those with HeFH or HoFH can present with tendon xanthomas, xanthelasma, arcus cornealis, premature ASCVD, and/or aortic valve stenosis, although each of these clinical findings tends to be more severe with earlier onset in persons with HoFH, for which such findings may be present in childhood.⁴⁵⁻⁴⁹

Isolated elevated TG levels

Hypertriglyceridemia is a complex disorder attributed to the combination of monogenic or polygenic influences and secondary factors, including acquired conditions and lifestyle behaviors, in most individuals.^{32,51-53} Therefore, in the initial workup of a patient with hypertriglyceridemia, contributing factors should be identified and managed. These include medical conditions (e.g., poorly controlled diabetes, hypothyroidism), medications (e.g., estrogens, beta-blockers, thiazide diuretics, corticosteroids), metabolic disorders (e.g., overweight/obesity, metabolic syndrome/insulin

resistance), and unhealthy dietary/lifestyle practices (e.g., excess alcohol and calorie intake, high intakes of added sugars and refined starches, and physical inactivity).^{3,32,52,53}

Most individuals with hypertriglyceridemia have mild-to-moderate TG elevations; however, some experience severe elevations with the presence of chylomicronemia, which increases the risk for pancreatitis, as well as other symptoms of chylomicronemia syndrome. Categories of hypertriglyceridemia are shown in Table 4.

Pathophysiology of mild-to-moderate hypertriglyceridemia

Fig. 2 shows key elements of TG metabolism. TGs in plasma are transported principally by VLDL particles, which enter circulation from the liver, and chylomicron particles (and their remnants), which enter circulation from the intestines via the thoracic duct. TG clearance is facilitated by lipoprotein lipase (LPL), which catalyzes hydrolysis of TGs from both VLDL and chylomicron particles, particularly in adipose tissue, skeletal muscle, and cardiac muscle. Several proteins are involved in the regulation of LPL activity, including Apo C2, which activates LPL, and Apo C3, which

Table 4 Categories of hypertriglyceridemia, chylomicronemia syndrome, and key causes of chylomicronemia syndrome ^{32,51-53}	
Hypertriglyceridemia Type	Definition
Persistent hypertriglyceridemia	Fasting TG level ≥150 mg/dL following: • A minimum of 4 to 12 weeks of lifestyle intervention; or • A stable dose of maximally tolerated statin (when indicated); or • Evaluation and management of secondary causes of hypertriglyceridemia
Mild-to-moderate hypertriglyceridemia	Fasting TG level ≥150 mg/dL or non-fasting TGs ≥175 mg/dL and <500 mg/dL
Severe hypertriglyceridemia	Fasting TG level ≥500 mg/dL and especially ≥1000 mg/dL with chylomicronemia, which increases risk for acute pancreatitis
Chylomicronemia syndrome	A syndrome that includes severe TG elevation (typically ≥1500 mg/dL), persistent chylomicronemia in the fasting state, and may include a markedly increased risk for acute pancreatitis, dyspnea, abdominal pain, eruptive xanthomas, lipemia retinalis, mental confusion, memory loss, and flushing with minimal alcohol intake
Main causes of severe hypertriglyceridemia with chylomicronemia	
Multifactorial chylomicronemia (formerly type V hyperlipoproteinemia)	The most common form of chylomicronemia occurs when secondary factors combine with genetic predisposition, leading to a markedly elevated circulating TG level with chylomicronemia
Familial chylomicronemia syndrome (formerly type I hyperlipoproteinemia)	A rare genetic disorder (prevalence ~1 per million) caused by mutations in genes for LPL, or proteins related to LPL regulation, leading to severe impairment of LPL activity and chylomicronemia; clinical findings often present early in life
Familial or acquired lipodystrophies	A rare group of disorders (prevalence ~1.5 to 4.5 per million) characterized by partial loss (in severe cases, generalized loss), or abnormal distribution, of adipose tissue, typically associated with marked insulin resistance and ectopic fat deposition

Abbreviations: LPL=lipoprotein lipase; TG=triglyceride

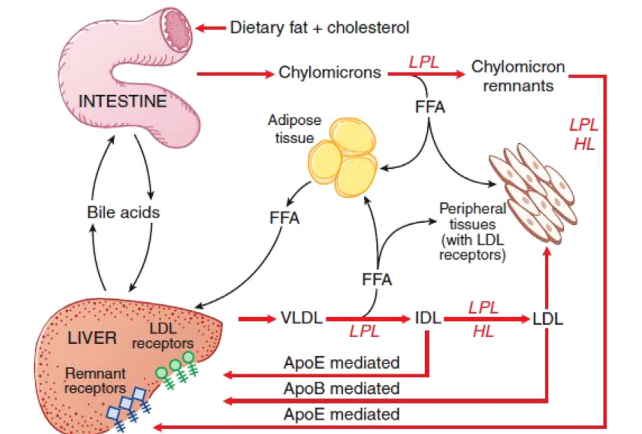


Fig. 2 Key elements of TG metabolism⁵⁷ This figure summarizes the major pathways for metabolism of chylomicrons synthesized in the intestine and VLDL synthesized in the liver.
Abbreviations: Apo B=apolipoprotein B; Apo E=apolipoprotein E; FFA=free fatty acids; HDL=high-density lipoprotein; HL=hepatic lipase, IDL=intermediate-density lipoprotein; LDL=low-density lipoprotein; LPL=lipoprotein lipase; TG=triglyceride; VLDL=very-low-density lipoprotein.

inhibits LPL. Apo C3 also interferes with hepatic uptake of VLDL and chylomicron remnant particles, contributing to impairment of TG clearance. Other LPL modulator proteins include Apo A5 (stimulates LPL), as well as ANGPTL3 and 4 (which inhibit LPL).^{32,51,52,54}

In mild-to-moderate hypertriglyceridemia, there is typically overproduction of VLDL-TG by the liver. This process

is often driven by excessive exposure of hepatocytes to free fatty acids, which can result from expanded adipose tissue stores, especially abdominal and visceral adipose tissue, as well as insulin resistance, which results in impaired suppression of free fatty acid release from adipose tissue.^{32,52,54,55} When VLDL-TG production is high, there is increased synthesis and release of VLDL particles that are enriched with Apo C3 and are Apo E deficient.⁵⁶ This results in impairment of TG clearance by LPL due to inhibition of LPL activity by Apo C3, and reduced uptake of VLDL and chylomicron remnants by the liver, a process that is partially dependent on Apo E and that is also inhibited by Apo C3. Thus, in most patients with persistent hypertriglyceridemia, both increased VLDL-TG output by the liver and impaired TG clearance are contributing factors.^{32,42,52,54}

Pathophysiology of severe hypertriglyceridemia and chylomicronemia

The clearance of TGs saturates at plasma concentrations of ~500-700 mg/dL.⁵⁸ When the TG level exceeds the saturation threshold, TG from new VLDL and chylomicron particles entering the circulation is removed very slowly, which can result in a marked elevation in circulating TGs and persistent chylomicronemia in the fasting state.⁵⁸ Chylomicronemia is not routinely measured in clinical settings because it requires specialized testing procedures.⁵⁹ It is reasonable to assume that some chylomicron particles are present in the fasting state in patients with TG concentrations ≥750 mg/dL and that significant chylomicronemia is present when the TG concentration exceeds 1000 mg/dL.^{51,60} The main risk from

chylomicronemia is the development of acute pancreatitis, the risk of which increases with the degrees of TG elevation and chylomicronemia.^{61,62} At TG levels >1770 mg/dL, the risk of acute pancreatitis is ~4% per year.⁶³ Pancreatitis is potentially life-threatening and, especially if repeated bouts are experienced, can lead to pancreatic insufficiency or chronic pancreatitis.⁶⁴

As summarized in Table 4, the three main causes of chylomicronemia are multifactorial chylomicronemia syndrome (MCS), familial chylomicronemia syndrome (FCS), and familial or acquired lipodystrophies. MCS is by far the most common of the three main causes of chylomicronemia and is associated with heterozygous variants in TG genes or the accumulation of multiple small-effect, single-nucleotide polymorphisms.⁶⁵ MCS may initially manifest as mild or moderate TG elevation that becomes more severe in the presence of contributing or acquired factors, e.g., overweight or obesity, insulin resistance, hyperglycemia, excessive consumption of simple sugars and/or alcohol, physical inactivity, and medications that may increase TG levels.^{51,66}

FCS is a rare disorder resulting from homozygous or compound heterozygous variants in one or more of five genes related to LPL function, specifically the *LPL* (60-80% of cases), *APOC2*, *GPIHBP1*, *APOA5*, and *LMF1* genes.^{66,67} In FCS, there is severe impairment of LPL activity resulting in markedly reduced hydrolysis of TG in chylomicron and VLDL particles and, consequently, long residence times for these lipoprotein particles in plasma, because some removal of TG is necessary before such particles can be effectively removed from circulation by the liver.⁵¹

Inherited lipodystrophies are rare autosomal recessive and dominant disorders characterized by loss or abnormal distribution of adipose tissue, although acquired forms also exist.^{52,68} Lipodystrophies are generally accompanied by insulin resistance and increased flux of free fatty acids, particularly postprandially, resulting in excessive VLDL-TG synthesis and secretion. As with other forms of hypertriglyceridemia, secondary factors contribute to the severity of the resulting hypertriglyceridemia.

Elevation in LDL-C and TG levels (mixed dyslipidemia)

The term mixed dyslipidemia is used herein to denote concurrent elevations in TG and LDL-C concentrations. Familial combined hyperlipidemia is the most common type of mixed dyslipidemia.⁶⁹ Mixed dyslipidemias are usually polygenic (concurrent polymorphisms associated with TG and LDL-C elevation) and often result from a combination of increased output of VLDL-TG (and to a lesser degree reduced TG clearance) and reduced clearance of LDL particles.^{32,69,70} When the TG concentration exceeds ~300-400 mg/dL, conversion of VLDL to LDL in circulation is typically impaired; therefore, it is common to encounter a rise in LDL-C when the TG level is lowered in such patients, reflecting a shift away from VLDL-C to LDL-C due to increased VLDL to LDL conversion.⁵⁶

Dysbetalipoproteinemia, formerly known as type III dyslipidemia, occurs when there is impaired uptake of TG-rich lipoprotein remnants by the liver combined with a second 'hit,' such as obesity or insulin resistance, which results in increased output and/or decreased clearance of VLDL-TG.^{71,72} The dysbetalipoproteinemia phenotype sometimes occurs when an individual is homozygous for a variant of Apo E (i.e., *APOE* $\epsilon 2/\epsilon 2$), which has reduced affinity for hepatic remnant receptors. The prevalence of the *APOE* $\epsilon 2/\epsilon 2$ genotype has been estimated at $\leq 1\%$ in cohorts with European ancestry and only a minority ($\leq 20\%$) of those individuals manifest the dysbetalipoproteinemia phenotype.⁷²⁻⁷⁴ However, the phenotype can manifest without the *APOE* $\epsilon 2/\epsilon 2$ genotype due to other contributors to poor hepatic remnant uptake (e.g., Apo E deficiency or variants that interfere with remnant receptor function) and dysbetalipoproteinemia has an estimated prevalence as high as ~2% using diagnostic methods that identify milder phenotypes.⁷⁴⁻⁷⁶ Because individuals with dysbetalipoproteinemia have decreased clearance of VLDL and chylomicron remnants, they will present with elevation in circulating levels of TG and cholesterol. This reflects increased levels of VLDL-C and VLDL-TG, as well as elevation in reported LDL-C, although this is generally the IDL-C portion of calculated LDL-C (i.e., an increased ratio of IDL-C to "true" LDL-C).⁵⁶ Because "true" LDL-C levels are generally not elevated but may appear so due to IDL-C elevation, the measurement of Apo B can be helpful for the diagnosis of dysbetalipoproteinemia.⁷⁷ Apo B will generally be low to normal in dysbetalipoproteinemia, despite non-HDL-C elevation.⁷⁷ In persons with dysbetalipoproteinemia who have more severe TG elevation, chylomicronemia may be present in the fasting state.^{51,60}

Nutrition interventions for the management of isolated elevation of LDL-C

Modifying intakes of cholesterol-raising dietary components

Several dietary components have been shown in RCTs to raise or lower levels of cholesterol carried by atherogenic particles (i.e., LDL-C and non-HDL-C). For patients with isolated LDL-C elevation, the objectives are to minimize cholesterol-raising dietary factors and increase intakes of dietary components that lower cholesterol within the context of a healthy dietary pattern. Dietary factors that increase LDL-C include SFAs (mainly 12-16 carbon SFAs), TFAs, and dietary cholesterol.⁷⁸⁻⁸¹ Excess adiposity and positive energy balance can also raise LDL-C.⁸² Dietary factors that lower atherogenic particles include unsaturated fatty acids,⁷⁸⁻⁸⁰ proteins (especially plant proteins),⁸³ viscous dietary fibers, and plant sterols/stanols (PSs).² The effects of each of these dietary components will be reviewed briefly.

Table 5 Predicted effects of isocaloric replacement of 5% of energy from SFAs with PUFAs, MUFAs, and carbohydrates on LDL-C, non-HDL-C, and Apo B based on results from 74 controlled feeding trials⁷⁹

Macronutrient	LDL-C (mg/dL)	Non-HDL-C (mg/dL)	Apo B (mg/dL)
PUFAs	-10.5	-11.0	-5.0
MUFAs	-8.0	-8.5	-4.0
Carbohydrates	-6.5	-6.0	-2.0

Abbreviations: Apo B=apolipoprotein B; HDL-C=high-density lipoprotein cholesterol; LDL-C=low-density lipoprotein cholesterol; MUFAs=monounsaturated fatty acids; PUFAs=polyunsaturated fatty acids; SFAs=saturated fatty acids.

Dietary saturated and unsaturated fatty acids

Dietary recommendations from health authorities and scientific organizations in the U.S. and internationally emphasize reducing dietary SFA intake and replacement with other nutrients, particularly unsaturated fatty acids, with an emphasis on polyunsaturated fatty acids (PUFAs).^{4-8,19} Table 5 shows the predicted effects on LDL-C and non-HDL-C of isocaloric replacement of 5% of energy from SFAs with PUFAs, monounsaturated fatty acids (MUFAs), and carbohydrates, based on a meta-regression analysis of controlled feeding trials.⁷⁹ Substitution with PUFAs produces the largest reductions, followed by MUFAs, and then carbohydrates. Although dietary recommendations emphasize reducing SFA intake as a class, it should be noted that not all SFAs are equally potent with regard to increasing LDL-C. It is primarily lauric, myristic, and palmitic acids (12-16 carbons) that raise levels of LDL-C and Apo B.⁷⁹ These are most abundant in butter and full-fat dairy products, processed and high-fat meats, and tropical oils, i.e., coconut, palm, and palm kernel oils.^{19,84}

Like SFAs, TFA consumption increases levels of atherogenic particles, with each 1% of energy from TFAs raising LDL-C by ~1.5 mg/dL compared with carbohydrates and, to a somewhat greater extent, compared with MUFAs and PUFAs.⁷⁸ The average intake of TFAs in the U.S. had been declining since the mid-1990s.⁸⁵ In 2015, the U.S. Food and Drug Administration (FDA) ruled that partially hydrogenated oils, the main source of industrially produced TFAs in the food supply, are not “generally recognized as safe,” and food manufacturers are no longer allowed to use partially hydrogenated oils, which has likely further reduced the intake of TFAs in the average American diet.⁸⁶ It is prudent for consumers to read the nutrition information on food labels to identify products with high levels of SFAs, any TFAs, and/or a high proportion of SFAs relative to total fat. Generally, it is desirable for foods to have 0 g of TFAs and <50% of total fat as SFAs (g of SFAs <50% of total g of fat per serving).

Dietary proteins

Compared to investigations of the effects of dietary fatty acids, fewer studies have evaluated the effects of dietary proteins on LDL-C and non-HDL-C levels. One such study is OmniHeart, a well-designed, randomized, 3-period, crossover, controlled feeding study (all food was provided), that evaluated the effects of three variations of the DASH

dietary pattern on lipoprotein lipids and blood pressure in participants with pre-hypertension or stage 1 hypertension.⁸³ After baseline measurements were collected with participants consuming their habitual dietary intake (average American diet), the three dietary interventions, each with a 6-week duration, were compared in a randomized order. All three DASH diet interventions were low in SFA (~6% total daily energy [TDE]). The macronutrient compositions of the diets were as follows:

- 1) a higher-carbohydrate diet with 58% TDE from carbohydrate, 27% from fat, and 15% from protein (the traditional DASH diet);
- 2) a higher protein DASH diet where 10% TDE from carbohydrate was replaced with mixed-source protein (~half from plant sources); and
- 3) a higher unsaturated fat DASH diet where 10% TDE from carbohydrate was replaced with unsaturated fatty acids (8% from MUFAs and 2% from PUFAs).

LDL-C and non-HDL-C levels were significantly reduced during all three dietary interventions compared with habitual intake (Table 6). However, reductions were greater with the higher protein and higher unsaturated fat interventions.

Results from meta-analyses of RCTs suggest that, when substituted for animal proteins, plant proteins from a variety of sources (~30 g/day or ~5% of TDE) reduce LDL-C and non-HDL-C concentrations (~7-9 mg/dL or 4%).^{87,88} Additionally, foods high in plant proteins, e.g., nuts, seeds, and legumes, contain other components that may enhance LDL-C-lowering, such as unsaturated fatty acids, viscous fibers, and naturally occurring PSs. Thus, in terms of dietary components that can reduce LDL-C, unsaturated fatty acids, primarily PUFAs, and proteins, particularly plant proteins, may be used to replace SFAs and refined carbohydrates (starches and added sugars) to elicit favorable effects on LDL-C and non-HDL-C levels.

Dietary cholesterol

The effects of dietary cholesterol on LDL-C levels have been evaluated in numerous studies. An increase of 100 mg/day in dietary cholesterol has been reported to increase LDL-C by ~1.5-2.0 mg/dL.^{81,89-91} However, the relationship may be non-linear⁸¹ and there appears to be substantial inter-individual variation in the sensitivity to dietary cholesterol, with some individuals experiencing little or no

Table 6 Changes from baseline lipoprotein lipid levels for the dietary interventions in OmniHeart⁸³

Lipoprotein Lipid	Dietary Intervention (% TDE from CHO/Protein/Fat)			
	Habitual BL Diet	High-CHO Diet (58/15/27)	Higher Protein Diet (48/25/27)	Higher UFA Diet (48/15/37)
	Mean, mg/dL	Mean Change from BL, mg/dL		
LDL-C	129	-11.6	-14.2	-13.1
Non-HDL-C	154	-11.0	-17.3	-15.1
TG	102	0.1	-16.4	-9.3

Note: Bold font indicates statistical significance relative to baseline ($p < 0.05$).
Abbreviations: BL=baseline; CHO=carbohydrate; HDL-C=high-density lipoprotein cholesterol; LDL-C=low-density lipoprotein cholesterol; TDE=today daily energy; TG=triglyceride; UFA=unsaturated fatty acids.

cholesterol-raising effect and others showing substantial elevations.⁹²⁻⁹⁴ In 2020, the AHA published a Science Advisory on dietary cholesterol and CV risk in which the authors reviewed the evidence from observational and intervention studies on the relationships between dietary cholesterol, lipoprotein lipids, and CVD risk.⁹⁵ The authors concluded that, within the range of dietary cholesterol and fats available from common food sources in the U.S., lowering the ratio of SFAs to unsaturated fatty acids has a greater potential impact on reducing LDL-C and non-HDL-C than reducing dietary cholesterol intake. Healthy dietary patterns are generally low in dietary cholesterol (100-300 mg/day) and the cholesterol intake in the average U.S. diet is <300 mg/day. The recommendation in this NLA Clinical Perspective for those with elevated LDL-C is consistent with that in the AHA Science Advisory, which is to consume a healthy dietary pattern that will inherently contain relatively low amounts of dietary cholesterol and SFAs.

Dietary adjuncts

If nutrition interventions and lifestyle therapies do not result in lowering LDL-C and non-HDL-C sufficiently, healthcare professionals may consider recommending dietary adjuncts to enhance reductions. The dietary adjuncts for which the most clear and consistent evidence exists for lowering LDL-C are viscous dietary fibers and PSs.

Viscous dietary fibers

Dietary fibers are plant substances that are resistant to digestion in the small intestine.⁹⁶ Fibers are classified in various ways, including water solubility, viscosity, and fermentability by colonic microbiota. Viscous fibers, including pectins, gums, mucilages, and some hemicelluloses, have gelling properties in the gastrointestinal tract, and their consumption has been associated with reductions in LDL-C, non-HDL-C, and Apo B.^{97,98} The main proposed mechanisms for the cholesterol-lowering effects of viscous fibers relate to trapping of cholesterol and bile acids in the small intestine, resulting in reduced absorption/reabsorption.⁹⁸ The net result of these actions is to reduce hepatic cholesterol content, resulting in up-regulation of hepatic LDL receptors that

remove Apo B-containing lipoproteins from the blood, thus lowering LDL-C and Apo B levels.⁹⁸

Commonly consumed foods high in viscous fibers include oats, barley, legumes (e.g., lentils, lima beans, kidney beans), and chia seeds, as well as fruits, including apples, peaches, and pears, and vegetables, including asparagus, Brussels sprouts, and sweet potatoes.⁹⁹ Supplemental forms of viscous fibers are available as fiber laxative products (e.g., those that contain psyllium husk and methylcellulose). Because psyllium seed husk (and other viscous fiber sources) may affect the absorption of some medications, it is best taken 2 hours before or 2 hours after medications.¹⁰⁰

A meta-analysis of 66 RCTs that examined the effects of viscous fibers from oat products (beta-glucan), psyllium, pectin, and guar gum demonstrated statistically significant reductions in LDL-C for each of the four fiber types, with no significant changes observed for TG or HDL-C concentrations. The pooled effects for all fibers per gram of daily consumption for all studies was a reduction of 1.1 mg/dL for LDL-C.¹⁰¹ These results were confirmed in a recent larger meta-analysis (165 RCTs).¹⁰² Accordingly, a viscous fiber intake in the range of 5-10 g/day would be expected to lower mean LDL-C levels by 5.5 to 11.0 mg/dL, or 4.6% to 9.2% with a pre-use LDL-C level of 120 mg/dL.

Plant sterols/stanols

PSs, also called phytosterols, which is a term used to refer to both plant sterols and plant stanols (e.g., sitosterol and sitostanol), are compounds that occur naturally in foods of plant origin, particularly vegetable oils, nuts, seeds, and grain products. The typical Western diet provides approximately 200-400 mg/day, and the highest quantity of PSs is consumed in a vegan diet (500-600 mg/day).^{2,103,104} Supplemental PS is available in a variety of functional foods,¹⁰⁵ although the availability of these products changes frequently. PS dietary supplements are also available for which consumers should be educated to purchase supplements with the “USP Verified Mark.” This ensures that the supplement has been produced using the U.S. FDA’s current Good Manufacturing Practices, contains the ingredients listed on the label, does not contain harmful substances, and will dissolve once taken.¹⁰⁶ A patient education handout on PSs is avail-

able on the NLA Clinician's Lifestyle Modification Toolbox webpage (www.lipid.org/clmt).

PSs lower LDL-C by reducing intestinal cholesterol absorption. This occurs because they compete with cholesterol for limited space in mixed micelles and, after absorption into the enterocyte, the increased sterol content triggers an up-regulation of adenosine triphosphate-binding cassette (ABC) transporters G5 and G8 that export sterols (including cholesterol) out of the enterocytes and into the intestinal lumen.¹⁰⁷ The resulting reduction in hepatic cholesterol content triggers up-regulation of hepatic LDL receptors that remove Apo B-containing lipoproteins from the blood, thus lowering the concentrations of LDL-C and Apo B.¹⁰⁷

Meta-analyses have shown reductions of 5-8% in LDL-C at an intake of ~2 g/day of PSs, with a dose-response effect that plateaus at an intake of ~3 g/day.¹⁰⁸⁻¹¹⁰ In 2014, the European Atherosclerosis Society Consensus Panel on Phytosterols reviewed the evidence on the LDL-C-lowering effect of PSs and found that 2 g/day lowered LDL-C levels by up to ~10%. The European Atherosclerosis Society Consensus Panel recommended functional foods with added PSs for a variety of patients with elevated LDL-C levels, including patients at low- to intermediate-risk for CVD with elevated LDL-C levels who may not require pharmacotherapy, as an adjunct for patients on lipid-lowering pharmacotherapy who have not yet attained their therapeutic objective for LDL-C, and patients with FH.¹⁰³ PS products generally have the greatest LDL-C-lowering effect when consumed with meals.^{104,108} The European Atherosclerosis Society Consensus Panel acknowledged that there were no RCT data with CV outcomes that established clinical benefit from the use of PSs.¹⁰³

Sitosterolemia (phytosterolemia) is a rare autosomal recessive disorder caused by biallelic loss-of-function genetic variants in either ABCG5 or ABCG8 genes, which leads to impaired elimination of PSs and hyperabsorption of PSs and cholesterol, resulting in markedly increased levels in blood and tissues and can contribute to development of atherosclerosis and xanthomas.¹¹¹ Patients with sitosterolemia should not use PS supplements or PS-fortified foods¹¹² and usually respond well to a diet low in PSs in conjunction with ezetimibe.^{111,113} Individuals with sitosterolemia may have LDL-C levels that range from normal to severely elevated, of which the latter may result in a misdiagnosis of FH.¹¹³ Sitosterolemia should be suspected in patients with LDL-C ≥ 190 mg/dL, who respond with a marked decrease in LDL-C levels with dietary restriction of PSs and/or ezetimibe use, or an increase in LDL-C with use of a PS supplement.^{111,113}

Other dietary supplements

The use of dietary supplements (also called nutraceuticals) for lipid-lowering has increased in recent years.^{114,115} However, the evidence to support the safety and efficacy for LDL-C lowering for dietary supplements other than viscous fibers and PSs is limited due to small numbers of studies, many with limited sample sizes, and concerns about risk of bias.^{114,116}

Red yeast rice (RYR), a dietary supplement promoted for LDL-C lowering, is produced via fermenting yeast (*Monascus purpureus*) grown on rice. During fermentation, several compounds are produced, including various monacolins (e.g., monacolins J, K, L, M, and X).^{117,118} Monacolin K is structurally identical to lovastatin and inhibits hydroxymethylglutaryl-coenzyme A (HMG-CoA) reductase.^{117,118} Other monacolins also have varying levels of HMG-CoA reductase inhibitory activity. Although meta-analyses of RCTs have shown significant reductions in LDL-C levels with doses generally between 1200 to 2400 mg/day of RYR extract,^{116,119-121} considerable variability exists in the LDL-C lowering response, which may be related to differences in monacolin content among brands and even between manufacturing runs of the same brand.¹¹⁷ A 2017 study analyzed 28 brands of RYR supplements sold in the U.S. to determine the monacolin K content. The results showed no detectable monacolin K in two brands. Of the other 26 brands, the monacolin K content ranged from 0.09 mg to 5.48 mg per 1200 mg of RYR.¹²² RYR products with any amount of monacolin K content above trace amounts are considered unapproved new drugs by the U.S. FDA.¹²³ However, the amount of monacolin K is not typically available on RYR product labels; therefore, it is impossible for healthcare professionals and consumers to know the amount of monacolin K in each product.¹²³ Furthermore, contaminants, such as citrinin, a nephrotoxin and carcinogen, have been found in marketed RYR.^{117,118}

Because of the paucity of high-quality evidence for many dietary supplements marketed for LDL-C lowering and the concerns about RYR discussed above, the authors of this NLA Clinical Perspective did not feel there was sufficient evidence to make a recommendation for the use of dietary supplements other than PSs and viscous fibers for LDL-C lowering.

Body weight and fat loss

Results from systematic reviews of RCTs indicate that weight loss of at least 2.5 kg, or 3% of body weight, is associated with clinically meaningful changes in ASCVD risk markers, including improvements in both LDL-C and TG levels.^{82,124} A weight loss of 5-8 kg that is sustained results in a mean LDL-C reduction of approximately 5 mg/dL (4.2% for a starting LDL-C level of 120 mg/dL).¹²⁴ However, healthcare professionals should be aware that there is marked variation in the lipoprotein lipid responses to weight loss. Additionally, all of the major lipoprotein lipid variables (LDL-C, VLDL-C, HDL-C, and TG) typically decrease during active weight loss.^{82,124} LDL-C levels decline after weight loss and stabilization at a lower body weight across the age spectrum, but the response tends to be larger in younger participants (<40 y) and may be blunted in older individuals (≥ 60 y).^{125,126} Favorable changes in lipoprotein lipids are unlikely to be sustained unless a reduced weight is maintained. Adverse changes in lipids associated with weight regain may result from changes in body composition,

as well as changes in dietary intake that occur with higher energy consumption and positive energy balance, particularly increased consumption of SFAs.⁸²

Much research has been conducted to identify the most effective dietary intervention for promoting weight loss and weight loss maintenance. The totality of the evidence suggests that reduced adiposity can be achieved with a variety of dietary interventions that result in reduced daily caloric intake.^{124,127-130} Despite the widely held belief that macronutrient composition is a key factor, the available evidence supports the view that dietary intervention adherence is the major predictor of weight loss success.^{124,131-133}

Comprehensive intervention programs with a multidisciplinary team that include structured nutrition and physical activity interventions and behavior counseling have produced the best results with regard to reducing adiposity and maintaining weight loss^{4,5,8,124,130,133} and should be made available to patients with overweight or obesity, particularly those with ASCVD or comorbidities, such as metabolic syndrome, diabetes mellitus, non-alcoholic steatohepatitis, or heart failure.¹³³ Such programs generally produce mean reductions in the range of 5-10% of initial body weight¹²⁴. Based on the results of clinical studies, persons with overweight or obesity who receive high-intensity lifestyle interventions, including participation in ≥14 weight loss intervention visits over 6 months with a trained interventionist, have improved outcomes compared with those who do not.^{124,130,133} In addition, results from a systematic review and meta-analysis showed that, compared to controls, weight management interventions provided by RDNs resulted in significantly greater percent weight loss (-4.0%; 95% confidence interval [CI]: -5.3, -2.8; 22 RCTs; high certainty of evidence) and waist circumference (-3.5 cm; 95% CI: -4.4, -2.5; 33 RCTs; high certainty of evidence).¹³⁰ For patients with more severe obesity and/or multiple comorbidities, consideration may be given to weight loss pharmacotherapy or surgery (e.g., sleeve gastrectomy or gastric bypass).^{4,124,133} It should be noted that loss of body weight typically involves loss of both body fat and lean mass (~25% of weight loss is generally attributed to loss of lean mass).¹³⁴ Inclusion of resistance exercise and adequate protein intake in the weight loss intervention will minimize loss of lean tissue.^{128,133} Behaviors associated with successful maintenance of a reduced body weight include high levels (≥250 min per week) of physical activity (mainly walking); eating breakfast regularly; self-monitoring of weight; and maintaining consistent, calorie-controlled eating patterns across weekdays and weekends.¹³⁵

Obesity is a highly prevalent chronic disease that is associated with health risks and requires life-long care. When addressing weight with patients, healthcare professionals should be cognizant of the potential for weight stigmatization and eating disorder risk.^{136,137} A recent consensus statement by six leading U.S. organizations focused on obesity care acknowledged that weight bias and stigmatization directed at people with obesity contribute to poor health and reduced treatment success.¹³⁸

Table 7 Estimated cumulative LDL-C lowering with selected nutrition interventions*

Nutrition Intervention	Estimated LDL-C ↓
Replacement of 5% energy from SFAs with unsaturated fatty acids	5% to 10%
7.5 g/day viscous fiber	6% to 9%
2 g/day plant sterols/stanols	5% to 8%
30 g/day plant protein†	3% to 5%
Loss of 5% of body weight, if excess adiposity is present	3% to 5%
Total expected effect	22% to 37%

*The sum of the cumulative effects of the selected nutrition interventions are theoretical based on the results from separate RCTs. Results from dietary intervention studies of free-living participants in which a portfolio of dietary interventions has been applied have shown up to a 30% LDL-C reduction in highly controlled settings and sustained reductions in the range of 12-15% in free-living participants¹³⁹⁻¹⁴¹.

†As a replacement for animal protein and/or carbohydrates.
Abbreviations: LDL-C=low-density lipoprotein cholesterol; SFAs=saturated fatty acids.

Additive effects of nutrition interventions on LDL-C lowering

A rule of thumb for healthcare professionals for predicting the effects of the recommended nutrition interventions outlined above is that each may lower LDL-C (and non-HDL-C) by ~5%. Combining nutrition interventions typically produces additive effects for the LDL-C response as illustrated in Table 7.

Combining any two of the interventions shown in Table 7 would be expected to reduce LDL-C by 6-19%, or 7-23 mg/dL, for a patient with a baseline LDL-C concentration of 120 mg/dL. The portfolio approach, which has been extensively studied in RCTs, combines a low-SFA diet with PSs, viscous fiber, soy protein, and almonds, and has been shown to reduce LDL-C by ~30% with controlled feeding; the reduction was less (~12-15%) when participants were free-living, but was maintained during follow-up periods of one year or longer.¹³⁹⁻¹⁴¹

RDNs provide individualized medical nutrition therapy and are invaluable for facilitating implementation of the nutrition interventions discussed above.^{31,33} The results of a meta-analysis of six RCTs demonstrated that, compared with control conditions (mainly usual care), medical nutrition therapy significantly decreased LDL-C levels (-11.6 mg/dL; 95% CI: -21.1, -2.0) with moderate certainty evidence.³⁰

Nutrition interventions for the management of isolated elevation of TG

The ACC has provided clinical recommendations for treating patients with elevated TG levels.⁵³ In agreement with the 2018 AHA/ACC/Multisociety Guideline on the Management of Blood Cholesterol,³ the ACC panel recom-

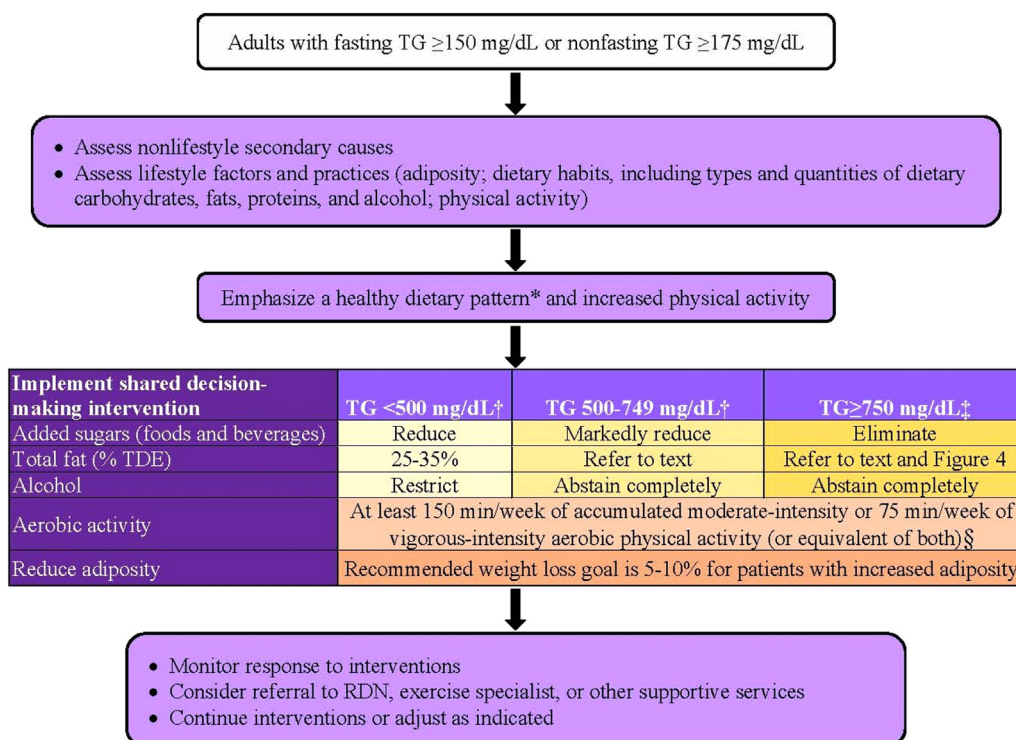


Fig. 3 Recommendations for lifestyle interventions in patients with increasing levels of triglycerides (Adapted from 53)

*Recommendations for a healthy dietary pattern emphasize vegetables, fruits, legumes (pulses), nuts and seeds, whole grains, fish/seafood (and other healthy proteins, such as low-fat dairy, low-fat poultry), non-tropical, liquid plant-based oils, and replacing SFAs with MUFAs and PUFAs. Recommendations also emphasize limiting red and processed meats, refined carbohydrates, foods and beverages with added sugars (sweets and sugar-sweetened beverages), sodium, dietary cholesterol, and avoiding TFAs.

[†]RDN referral advised.

[‡]RDN referral necessary.

[§]Although healthcare professionals should encourage patients to meet the guideline-recommended goals for physical activity, any amount of physical activity is likely beneficial in sedentary individuals and, therefore, should be encouraged to reduce cardiometabolic risk.

Abbreviations: MUFAs=monounsaturated fatty acids; PUFAs=polyunsaturated fatty acids; RDN=registered dietitian nutritionist; SFAs=saturated fatty acids; TDE=total daily energy; TFAs=*trans* fatty acids; TG=triglyceride

mended lifestyle intervention as the foundation for management of patients with hypertriglyceridemia.⁵³ The recommendations herein have been adapted from the ACC panel guidance.

There are commonalities in the nutrition and physical activity recommendations for all patients with hypertriglyceridemia, including limiting intakes of added sugars, refined starches, and alcohol; weight loss for patients with overweight or obesity; and glycemic management for patients with diabetes mellitus.^{32,52,53} Adequate physical activity is essential because it increases fat oxidation, enhances TG clearance, improves insulin sensitivity,¹⁴² and facilitates body weight management.^{124,133} Physical activity recommendations include ≥ 150 min/week of moderate-intensity activity (e.g., brisk walking), ≥ 75 min/week of vigorous-intensity activity (e.g., jogging or running), or the equivalent combination, plus 2-3 days/week of muscle strengthening exercises.^{4,5,7,8,133,143} Nutrition intervention recommendations differ in several aspects according to the severity and underlying causes of TG elevation (Figs. 3

and 4). Because many individuals with hypertriglyceridemia have underlying polygenic and acquired determinants, it is critical to individualize nutrition interventions. A key consideration is to monitor TG levels, especially early in the intervention, to assure that the recommended nutrition intervention has the desired effects. There is substantial inter-patient variability in response to nutrition interventions and some trial and error should be expected. In the presence of hypertriglyceridemia, especially when the TG level is ≥ 500 mg/dL, it is essential to obtain fasting baseline and follow-up TG levels to assess response.^{52,53}

For individualized management and to promote long-term adherence for patients with any category of hypertriglyceridemia, and especially in patients with severe TG elevation, nutritional counseling and follow-up/monitoring by an RDN is recommended.⁵³ The results of a systematic review of five RCTs showed that, compared to control conditions (mainly usual care), medical nutrition therapy provided by an RDN improved TG levels (-32.6 mg/dL; 95% CI: -57.8 , -7.3 , moderate certainty evidence).³⁰

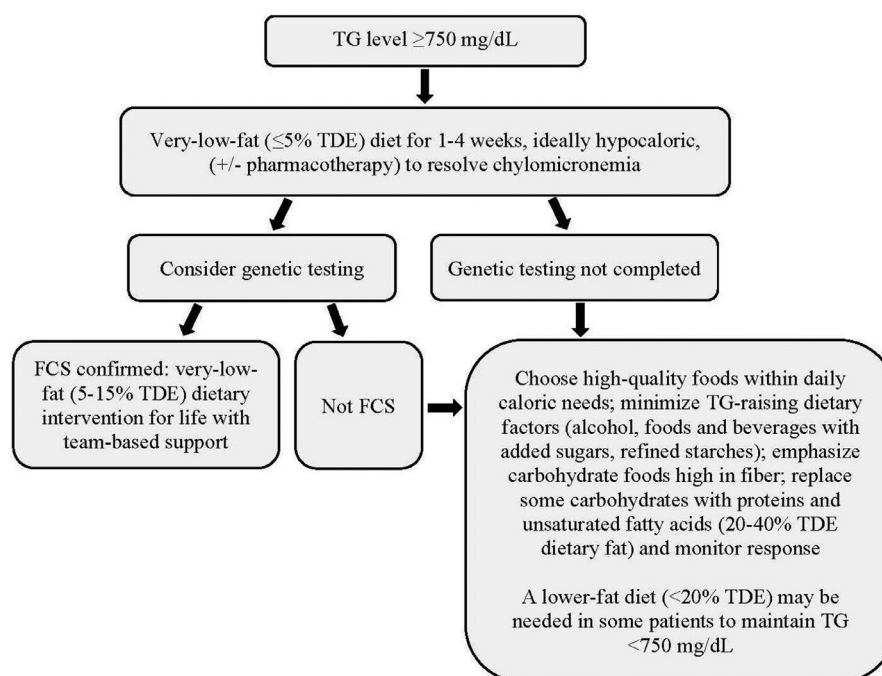


Fig. 4 Algorithm for nutrition interventions for severely elevated triglyceride levels (≥ 750 mg/dL)^{51,144}

Abbreviations: FCS=familial chylomicronemia syndrome; TDE=total daily energy; TG=triglyceride.

Nutrition interventions for mild-to-moderate hypertriglyceridemia (fasting level ≥ 150 to < 500 mg/dL)

Mild-to-moderate hypertriglyceridemia typically clusters with other cardiometabolic risk factors (e.g., abdominal obesity, hypertension, hyperglycemia, chronic inflammation, and metabolic syndrome), and is associated with increased risk for ASCVD.^{32,52,53} Some of the increase in ASCVD risk is likely attributable to elevated levels of atherogenic lipoproteins present with hypertriglyceridemia (e.g., TG-rich lipoprotein remnant particles; small, dense LDL particles resulting in an increased Apo B to LDL-C ratio), while a portion may be secondary to associated metabolic disturbances, such as insulin resistance, hyperglycemia, inflammation, and increased thrombogenicity.^{1,32,52-54} As discussed previously, factors contributing to hypertriglyceridemia (nutrition- and lifestyle-related, medical conditions, and medications) should be identified and managed. Fig. 3 summarizes the established lifestyle interventions for the management of mild-to-moderate hypertriglyceridemia and Table 8 summarizes the recommended nutrition interventions for the management of mild-to-moderate hypertriglyceridemia. TG reductions of 10-40% often occur with the implementation of the recommended lifestyle interventions, although highly responsive individuals may achieve reductions of $\geq 70\%$.^{32,53} Reducing adiposity in patients with overweight or obesity and intake of dietary sources of omega-3 fatty acids warrant more detailed discussion.

Body weight and adiposity management

Increased adiposity is a key driver of TG elevation and contributes to insulin resistance. For patients without overweight or obesity (BMI < 25 kg/m² or < 23 kg/m² for Asian Americans),¹⁴⁵ the focus should be on avoiding weight gain while maintaining healthy lifestyle habits. For patients with overweight or obesity, weight loss decreases TGs in a dose-dependent manner. Clinically meaningful reductions in TGs are observed with sustained weight loss of 3-5% of body weight.^{52,53,124,125,133} As discussed in the section on nutrition interventions for elevated LDL-C levels, reduced adiposity can be achieved with a variety of dietary interventions that result in reduced daily caloric intake.^{124,127-130,133} For patients with mild-to-moderate TG elevation with overweight or obesity, a reduced calorie dietary pattern that meets the nutrition recommendations in Table 8 and aligns with personal preferences and cultural considerations should be advised.

Omega-3 fatty acids

It is recommended in the Dietary Guidelines for Americans,¹⁹ as well as guidelines and recommendations specific to ASCVD prevention,^{6,8,53,146} that all individuals consume ≥ 2 servings/week of fish or seafood (total of 8 oz/week). Vegans prefer not to consume fish, seafood, or fish oil supplements and may consider alternative sources of omega-3 fatty acids, such as long-chain fatty acid supplements from algal oil. For patients with mild-to-moderate TG elevation, intake of fatty fish, which is rich in the long-chain omega-3 fatty acids eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA) should be encouraged (e.g., salmon, halibut, albacore

Table 8 Features of healthy dietary patterns that meet nutrition recommendations for hypertriglyceridemia based on triglyceride elevation and presence of chylomicronemia^{*53,161}

Food Groups	Mild-to-Moderate TG Elevation (≥150 to <500 mg/dL) ^{**†}	Severe TG Elevation (≥500 to 749 mg/dL) ^{**†}	Severe TG Elevation (≥750 mg/dL) – presumed MCS ^{**‡}	Severe TG Elevation (≥750 mg/dL) – FCS ^{**‡}
Calories	Meet daily needs with minimally processed nutrient-dense foods	Meet daily needs with minimally processed nutrient-dense foods	Meet daily needs with minimally processed nutrient-dense foods	Meet daily needs with minimally processed nutrient-dense foods; gradually add MCT oil, as needed
Alcohol	Restrict; if choosing to drink: Men ≤2 drinks/day ^{***} Women ≤1 drink/day ^{***}	Abstain completely	Abstain completely	Abstain completely
Sugar-sweetened beverages	Restrict	Abstain completely	Abstain completely	Abstain completely
Fruits [#]	Mostly whole fruits, 2 servings/day ^a	Mostly whole fruits, 2 servings/day ^a	Mostly whole fruits, 2 servings/day ^a	Mostly whole fruits, limit to 1 serving/day ^a
Vegetables [#]	2 ½ cups/day ^b	2 ½ cups/day ^b	2 ½ cups/day ^b	2 ½ cups/day ^b
Grains [#]	Mostly fiber-rich whole grains, 4-6 servings/day ^c	Mostly fiber-rich whole grains, 4-6 servings/day ^c	Mostly fiber-rich whole grains, 4-6 servings/day ^c	Individualize ^c
Dairy [#]	Limit full-fat dairy and/or sugar-sweetened dairy; choose low-fat or fat-free unsweetened dairy or unsweetened dairy alternatives, 3 servings/day ^d	Limit full-fat dairy and/or sugar-sweetened dairy; choose low-fat or fat-free unsweetened dairy or unsweetened dairy alternatives, 3 servings/day ^d	Limit full-fat dairy and/or sugar-sweetened dairy; choose low-fat or fat-free unsweetened dairy or unsweetened dairy alternatives, 3 servings/day ^d	Eliminate full-fat dairy and/or sugar-sweetened dairy; choose fat-free unsweetened dairy or unsweetened dairy alternatives, 3 servings/day ^d
Protein sources [#]	5 ½ oz equivalents/day ^e	5 ½ oz equivalents/day ^e	5 ½ oz equivalents/day ^e	Individualize servings/day
Legumes (beans, soy), nuts, and seeds	Choose mostly plant protein sources	Choose mostly plant protein sources	Choose mostly plant protein sources; cautiously include nuts and seeds	Choose mostly plant protein sources; eliminate nuts and seeds due to fat content
Fish and seafood	Emphasize fatty fish, ≥2 servings/week (8 oz/week)	Emphasize fatty or lean fish, ≥2 servings/week (8 oz/week)	Emphasize fatty or lean fish, ≥2 servings/week (8 oz/week)	Emphasize lean fish, ≥2 servings/week (8 oz/week)
Poultry/meats	Choose lean cuts and avoid processed meats	Choose lean cuts and avoid processed meats	Choose lean cuts and avoid processed meats	Choose very lean cuts without skin and avoid processed meats

*Individualize the nutrition recommendations, including macronutrient distribution, calories, and number of servings from each food group based on patient needs, TG elevation, and TG response.

**Any individual with an elevated TG level has some degree of LPL impairment, which can range from minimal to severe. Most individuals with TGs ≥500-749 mg/dL do not have severe LPL impairment. Individuals with TGs persistently ≥750 mg/dL are more likely to have severe LPL impairment.

†RDN referral advised.

‡RDN referral necessary.

***If an individual chooses to drink, moderate consumption of alcohol should be emphasized. However, some individuals are more sensitive to the TG-raising effects of alcohol intake and may require restriction or abstinence. Additionally, alcohol reduction may facilitate weight loss in persons with overweight or obesity and will help reduce hepatic steatosis that commonly occurs in the context of hypertriglyceridemia.

[#]Number of servings per day is based on 2,000 calories/day and should be individualized.¹⁹

^a1 fruit serving = 1 medium fruit; 1 cup raw, cooked, canned in natural juice, or frozen with no added sugar; 1 cup 100% fruit juice; ½ cup dried fruit.

^b1 vegetable serving = 1 cup raw or cooked; 1 cup vegetable juice; 2 cups leafy salad greens; limit intake of starchy vegetables to no more than 2 servings/day.

^c1 grain serving = ½ cup cooked rice, pasta, or cereal; 1 oz dry pasta or rice; 1 medium (1 oz) slice bread, tortilla, or flatbread; 1 oz of ready-to-eat cereal (about 1 cup of flaked cereal).

^d1 dairy serving = 1 cup milk, yogurt, or fortified milk alternative; 1½ oz natural cheese, such as cheddar cheese or 2 oz of processed cheese.

^e1 ounce equivalent for protein = 1 oz lean meats, poultry, or seafood; 1 egg; ¼ cup cooked beans or tofu; 1 tbsp nut or seed butter; ½ oz nuts or seeds.

Abbreviations: FCS=familial chylomicronemia syndrome; LPL=lipoprotein lipase; MCS=multifactorial chylomicronemia syndrome; MCT=medium-chain triglyceride; RDN=registered dietitian nutritionist; PCS=primary chylomicronemia syndrome; TG=triglyceride.

tuna). Intake of two servings of fatty fish per week provides the equivalent of 450-500 mg/day of EPA + DHA.¹⁴⁷

EPA and DHA lower the circulating TG level in a dose-dependent manner, with reductions $\geq 15\%$ occurring at intakes of 2-4 g/day.¹⁴⁸ While omega-3 dietary supplements may be considered, caution is warranted because some supplements may be of poor quality and could contain contaminants and oxidation products, in addition to SFAs and cholesterol.¹⁴⁹ Prescription omega-3 products containing EPA alone or EPA + DHA are available and have U.S. FDA-approved indications for treatment of severe hypertriglyceridemia.^{148,150} Products containing icosapent ethyl (EPA ethyl esters) also have an approved indication for reduction of ASCVD risk in patients with hypertriglyceridemia (fasting TG ≥ 150 mg/dL) who are at increased risk for an ASCVD event due to established ASCVD or type 2 diabetes mellitus.^{148,150}

Nutrition interventions for severe hypertriglyceridemia (500 mg to 749 mg/dL)

For patients with fasting TG levels 500-749 mg/dL, some elements of nutrition interventions are the same as those recommended for mild-to-moderate TG elevation, i.e., limiting intakes of added sugars and refined starches, limiting or eliminating alcohol, participating in daily physical activity, and reducing adiposity (Fig. 3). However, there is more variability in underlying causes for patients with TG levels between 500-749 mg/dL, thus implementing the appropriate nutrition interventions involves monitoring the TG response and assessing the individual's tolerance to the amount of total fat consumption.

It is a widely held belief that all patients with severe hypertriglyceridemia require a very-low-fat diet; however, there is a paucity of evidence to support this view. Limited evidence is available from RCTs of nutrition interventions in severe hypertriglyceridemia. The studies that have been completed generally have had small numbers of participants and have not always fully controlled possible confounding issues, such as negative energy balance and weight loss.^{42,151-153} The results of nutrition intervention studies that included participants with hypertriglyceridemia suggest that there is substantial variability in the TG response and that the optimal dietary approach likely varies according to the balance of the underlying metabolic drivers, which include rates of entry of TG into the circulation from intestinally-derived chylomicron particles and hepatically-derived VLDL particles, rates of TG hydrolysis by LPL, and the rate of hepatic uptake of TG-rich lipoproteins.

For patients with FCS, a very low-fat (5-15% TDE) diet is needed long term (discussed further below) to minimize the production of chylomicron particles because LPL activity is severely impaired ($<25\%$ of normal).^{144,154} However, FCS is rare (~ 1 in 1,000,000)^{51,66} thus, most patients with severe hypertriglyceridemia will have less severe impairment of LPL activity and excessive VLDL-TG output and/or impaired clearance of TG-rich lipoprotein particles will also be

important drivers of TG elevation.^{32,54,58} For patients with fasting TG ≥ 750 mg/dL, it is reasonable to assume that some degree of chylomicronemia is present and the initial implementation of a very-low-fat diet ($\leq 5\%$ TDE), ideally at least mildly hypocaloric, will be helpful to reduce chylomicronemia. Subsequently, especially if the TG level has been lowered to <500 mg/dL, dietary fat intake may be cautiously liberalized with monitoring of the TG response. Some patients will respond to liberalization of dietary fat with a rise in TG level. Such individuals may require a lower fat diet ($<20\%$ TDE) to avoid TG elevation and chylomicronemia.¹⁵²

Factors that contribute to increased VLDL-TG output include excess adiposity, insulin resistance, hyperinsulinemia, hyperglycemia, positive energy balance, physical inactivity, consumption of added sugars and refined starches, and alcohol intake.^{32,155} For patients whose hypertriglyceridemia is driven primarily by excessive VLDL-TG output, a diet lower in carbohydrate will often produce the greatest degree of TG reduction.^{151,153,156} For such patients, a diet in which some dietary carbohydrate is replaced by unsaturated fatty acids and proteins may be preferred (e.g., 30-40% TDE fat, 15-25% TDE protein, and the remainder from carbohydrate).

A high intake of fructose from processed foods, fruit juices, and sugar-sweetened beverages has an especially detrimental effect on TG levels, particularly postprandially, due to its impact on metabolic pathways that enhance lipogenesis and TG synthesis.^{52,53,157,158} Fructose comprises $\sim 50\%$ of the mass of sucrose and high-fructose corn syrup and is a substantial component of some other commonly used sweeteners, such as honey, agave nectar, molasses, and raw sugar.⁵²

In summary, nutrition intervention for severe hypertriglyceridemia (500-749 mg/dL) should emphasize a healthy dietary pattern with greater emphasis on consumption of foods that are rich in healthy proteins, dietary fibers, and unsaturated fatty acids (including long-chain omega-3 fatty acids), such as nuts, seeds, legumes, fish/seafood, and non-tropical, liquid plant oils and should minimize or eliminate intakes of sweets, desserts, and dried fruits, fruit juices, sugar-sweetened beverages, and alcohol.^{32,52,53} As is the case for nutrition interventions for less severe elevations in TG levels, considerable interindividual variability should be expected and monitoring and readjustments may be necessary to determine what dietary approach will provide the best balance of patient preferences, adherence, and TG response.

Nutrition interventions for severe hypertriglyceridemia (≥ 750 mg/dL)

For patients with TG ≥ 750 mg/dL who may be presumed to have a significant degree of chylomicronemia in the fasting state, a very-low-fat ($\leq 5\%$ TDE) diet, ideally at least mildly hypocaloric, for 1-4 weeks is helpful initially, which may be combined with pharmacotherapy, to reduce TGs below 750 mg/dL.^{51,144,152} Restriction of dietary fat will reduce the production of new chylomicron particles.^{51,144,152} Negative energy balance will lower the circulating insulin

level and increase hepatic fat oxidation, and thus reduce hepatic VLDL-TG output.^{125,159,160} With less new TG entering the circulation from both the intestines as chylomicron-TGs and liver as VLDL-TGs, chylomicronemia will typically resolve.

The optimal maintenance diet will depend on the underlying metabolic disturbances. When TGs have decreased to <750 mg/dL, for those without known FCS, dietary fat intake may be cautiously liberalized with monitoring of the TG response.¹⁵³

Nutrition interventions for FCS and other types of severe LPL deficiency

Nutrition therapy for FCS, which also applies to other conditions that produce severe LPL deficiency, differs from other types of hypertriglyceridemia because of the need for marked dietary fat restriction long term to reduce chylomicronemia and prevent potentially life-threatening pancreatitis. Dietary guidance for FCS has been described by Williams et al.¹⁶¹ and includes:

- 1) Limit fat to <15-20 g/day (<10-15% TDE) with dietary fat spread across the day.
- 2) Meet recommendations for essential fatty acids, i.e., α -linolenic acid and linoleic acid.
- 3) Choose complex carbohydrate foods while limiting refined starches. Patients with FCS must avoid alcohol and foods and beverages with added sugars.
- 4) Supplement with fat-soluble vitamins and minerals, as needed.
- 5) Adjust calorie intake and physical activity for weight management.
- 6) If additional calories are needed, the diet may be supplemented with small amounts of medium-chain triglyceride (MCT) oil. Medical-grade MCT oil contains only the medium-chain fatty acids capric and caprylic acids and is not the same as coconut oil or some coconut oil-based MCT oils, which con-

tain lauric acid. Lauric acid is largely metabolized via a chylomicron-dependent pathway. Medical-grade MCT oil should be initiated in small amounts and gradually increased to tolerance. Guidance for the practical aspects of using MCT oil is available from the FCS Focus website (<https://fcsfocus.com/nutrition/nutrition-lifestyle-facts/>).

Additional nutrition recommendations and foods that can be emphasized for patients with and without FCS are summarized in Table 8. The dietary restrictions, while meeting energy needs and nutrient requirements, are challenging for FCS patients to follow. Consequently, it is essential that patients with FCS be referred to an RDN for medical nutrition therapy.^{53,161}

Nutrition interventions for elevated LDL-C and TG (mixed dyslipidemia)

Patients with elevations in both LDL-C and TGs will typically have reduced clearance of LDL particles concurrent with elevated VLDL-TG output and some degree of impairment in TG clearance.⁶⁹ A majority of such patients will have mild-to-moderate TG elevation, although some patients with more severe TG elevation initially may experience an increase in LDL-C as the TG concentration is lowered, resulting in mixed dyslipidemia⁵⁶ (Table 3). Nutrition interventions for such patients will consist of combining the strategies employed for lowering LDL-C and for reducing TGs discussed previously within the context of a healthy dietary pattern. A summary of these dietary strategies is shown in Table 9.

Conclusion/Summary

This Clinical Perspective describes recommended nutrition interventions for the management of elevations in LDL-C and TGs individually, and in combination. Various genetic determinants and other (acquired) factors contribute to the

Table 9 Summary of nutrition and lifestyle interventions to lower circulating levels of LDL-C and TG*

Interventions for Lowering LDL-C	Interventions for Lowering TG**
↓ SFAs, TFAs, and dietary cholesterol	↓ Added sugars and refined starches
↑ UFAs intake (5% TDE replacement for SFAs)	↓ Alcohol
↓ Body weight (5-10%), if overweight/obese	↓ Body weight (5-10%), if overweight/obese
↑ Protein, especially plant protein (3-5% TDE)	↑ Protein, especially plant protein (3-5% TDE)
↑ Viscous fiber intake (5-10 g/day)	↑ EPA+DHA intake (2-4 g/day)
↑ Plant stanols/sterols (2 g/day)	↑ Physical activity (≥ 150 minutes/week) [†]

*The dietary strategies summarized are within the context of an overall healthy lifestyle that includes a recommended dietary pattern, adequate physical activity, avoidance of tobacco products, adequate sleep quantity and quality, and psychosocial stress management.

**In addition to the interventions listed, individualize the macronutrient composition of the dietary pattern. Many patients will achieve TG reduction with dietary fat intake in the range of 20-40% TDE, whereas a lower fat diet (<20% TDE) may be needed in a small number of patients to maintain TG <750 mg/dL.

[†] ≥ 150 min/week of moderate-intensity activity (e.g., brisk walking), ≥ 75 min/week of vigorous-intensity activity (e.g., jogging or running), or the equivalent combination, plus 2-3 days/week of muscle strengthening exercises.^{4,5,7,8,143}

Abbreviations: DHA=docosahexaenoic acid; EPA=eicosapentaenoic acid; LDL-C=low-density lipoprotein cholesterol; SFAs=saturated fatty acids; TDE=total daily energy; TG=triglyceride; TFAs=trans fatty acids; UFAs=unsaturated fatty acids.

development of these dyslipidemias. Modifiable secondary factors (e.g., medications and medical conditions) that contribute to LDL-C and/or TG elevation should be identified and addressed. For reducing ASCVD risk, the main targets of nutrition interventions are LDL-C and non-HDL-C, which are generally surrogates for the concentration of circulating Apo B-containing atherogenic lipoprotein particles. Nutrition interventions for addressing dyslipidemias should be instituted within the context of a healthy dietary pattern and other lifestyle behaviors (adequate physical activity, abstinence from tobacco, sleep hygiene, and stress management) that promote CV health.

Nutrition interventions that lower elevated LDL-C levels include reducing cholesterol-raising dietary components (SFAs and cholesterol), increasing dietary components that lower LDL-C (unsaturated fatty acids, plant proteins, viscous fibers, and PSs), and reducing adiposity for patients with overweight or obesity. The portfolio approach combines a low-SFA diet with PSs, viscous fiber, soy protein, and almonds and has been shown to reduce LDL-C by ~30% with controlled feeding. The reduction was less (~12–15%) when participants were free-living but was maintained during follow-up periods of one year or longer. A reduction of this magnitude (18 mg/dL for a baseline LDL-C level of 120 mg/dL), if maintained over an extended period (decades), would be expected to lower major vascular event risk by ~30%.

Nutrition interventions for all patients with hypertriglyceridemia include restricting intakes of alcohol, added sugars, and refined starches. Additional lifestyle factors that are important for reducing TG levels are participating in daily physical activity and reducing adiposity in patients with overweight or obesity. Absolute or relative insulin deficiency can be a contributing factor for hypertriglyceridemia; therefore, pharmacological interventions to improve glycemic control should be instituted simultaneously with lifestyle interventions for reducing TG levels. Nutrition interventions for addressing concurrent elevations in LDL-C and TGs include a combination of the strategies described for lowering LDL-C and TG levels.

For patients with severe TG elevation (≥ 500 mg/dL), the primary therapeutic objective is to lower the TG concentration to reduce the risk for pancreatitis, the incidence of which increases with the degree of chylomicronemia. A secondary objective in patients with severe TG elevation is reducing ASCVD risk, which can occur in such patients with severe TG elevation who have elevated Apo B levels or dysbetalipoproteinemia. Patients with TG ≥ 750 mg/dL, especially if ≥ 1000 mg/dL, will generally have chylomicronemia in the fasting state. A very-low-fat, ideally at least mildly hypocaloric, diet for 1–4 weeks will reduce chylomicron formation and help to resolve chylomicronemia. The optimal diet for the longer term in such patients will vary and a degree of trial and error should be expected. Some patients will require a chronically low-fat diet (<20% TDE) long term to avoid chylomicronemia. Others may be able to consume a dietary pattern with a higher fat intake (20–40% TDE), af-

ter chylomicronemia has been cleared, without aggravating hypertriglyceridemia.

A multidisciplinary approach is recommended to facilitate patient success in making and sustaining dietary changes and the assistance of an RDN is highly recommended whenever feasible. This NLA Clinical Perspective has primarily focused on nutrition interventions, but other lifestyle and behavioral modifications for the reduction of ASCVD risk are also important and will be described in greater detail in future NLA Clinical Perspectives.

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CFK is an employee of Midwest Biomedical Research and has no additional conflicts of interest to report beyond those disclosed for KCM.

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